

Concept note on valence recognition for human face image data

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Abbreviations and definitions

Abbrev	Longform (= translation)
al.	aliī = others
Feb	february
cf.	cōnfer = compare
CNN	convolutional neural network
CUDA	Compute Unified Device Architecture
dict	dictionary
DLBDSEAIS02	Distance Learning Bachelor Data Science Elective Artificial Intelligence – course #2
Dr.	doctor
dvc	Data Version Control
e.g.	exemplī grātiā = for example
etc.	et cetera
FER	Facial Expression Recognition
float	floating-point number
gh	github
GPU	graphics processing unit
IU	International University of Applied Sciences
matplotlib	MATLAB, plot, and library
ML	machine learning
MTCNN	Multitask Cascaded Convolutional Networks
ndarray	N-dimensional array
p./pp.	page/pages
PIL	Python Imaging Library
sns	Samuel Norman Seaborn
str	string
ui	user interface
UML	Unified Modeling Language

1. Problem statement

We have been requested to build a tool that can detect the emotional state of a person while watching the company's advertisements. The goal is to measure viewer reactions to ad content by analyzing facial expressions captured in still images. We strive towards a **threefold valence categorization** — *negative*, *neutral*, and *positive* — which provides a clean signal for marketing analytics.

2. Envisioned product

The system receives an image containing a human face. Detects and localizes a face in the input image. Calculates the emotion probability distribution of the expression relative to seven standard emotions (cf. Zhao et al., 2023, p. 2758). Maps the distribution onto a 3-class valence scale based on a valence mapping (cf. [Appendix](#)). Returns a **valence label** (negative/neutral/positive), a **confidence score** for the prediction, plus the underlying (7-class) **emotion probability distribution** from which the valence was derived. 5. Validates classification quality on benchmark data with standard ML metrics.

3. Timeline and risks

Step	Task	Duration	Dependencies	Risk
0	Set up project	1 hour	Python, git, gh, dvc, uv	
1	Obtain reorganized FER-2013 data (Jangid, 2025)	1 day	Kaggle	
2	Explore & preprocess data	4 days	PyCharm, PIL	label noise
3	Set up GPU usage	2 days	CUDA, Lightning, PyTorch	Blackwell nightly build issues
4	Integrate pretrained model	6 days	CNN (per Bai et al., 2018) from HuggingFace, facenet-pytorch	extended research
5	Build valence mapping layer	1 day		
6	Implement validation module	4 days	scikit-learn	ambiguous ground truth
7	Build web ui for demo	5 days	gradio, gradio-client	connections
8	Create notebook presentation	5 days	JupyterLab, matplotlib, sns	getting lost in workflow details
9	Testing and documentation	4 days	pytest	unexpected results
10	Put into company system	4 days	AWS S3, Step Functions	skilled resource (?)

Bibliography

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- Zhao, R., Liu, T., Huang, Z., Lun, D. P. K., & Lam, K.-M. (2023). Spatial-Temporal Graphs Plus Transformers for Geometry-Guided Facial Expression Recognition. *IEEE Transactions on Affective Computing*, 14(4), 2751–2767. (173946068).
<https://doi.org/10.1109/TAFFC.2022.3181736>

APPENDIX

Emotion	Valence	Rationale
Angry	Negative	Clear negative affect
Disgust	Negative	Clear negative affect
Fear	Negative	Clear negative affect
Sad	Negative	Clear negative affect
Neutral	Neutral	No dominant affect
Happy	Positive	Clear positive affect
Surprise	Positive	In the context of advertisement viewing, surprise is positive

Figure 1: Base emotion ◀▶ valence mapping

System Architecture – UML Class Diagram

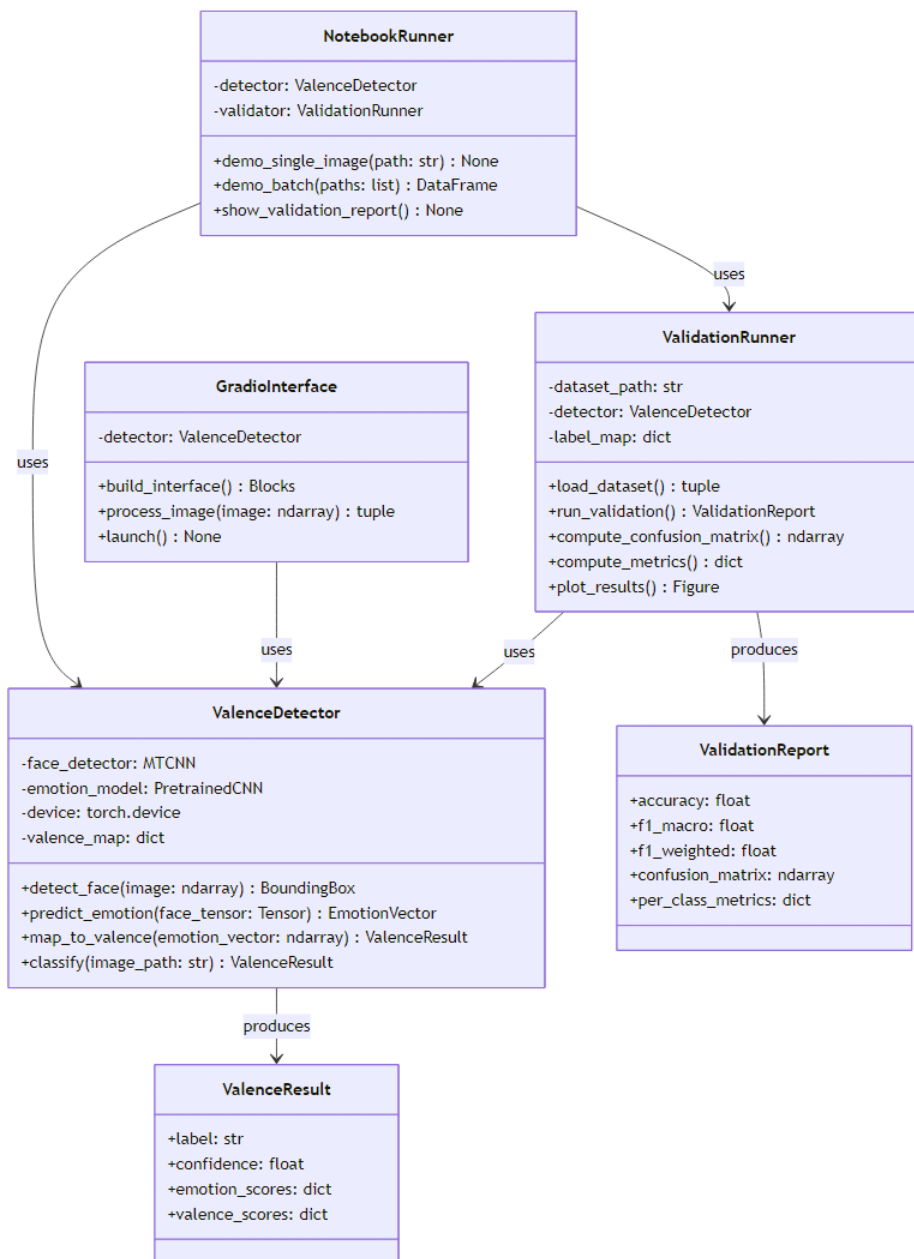


Figure 2: System architecture expressed in the form of a UML class diagram

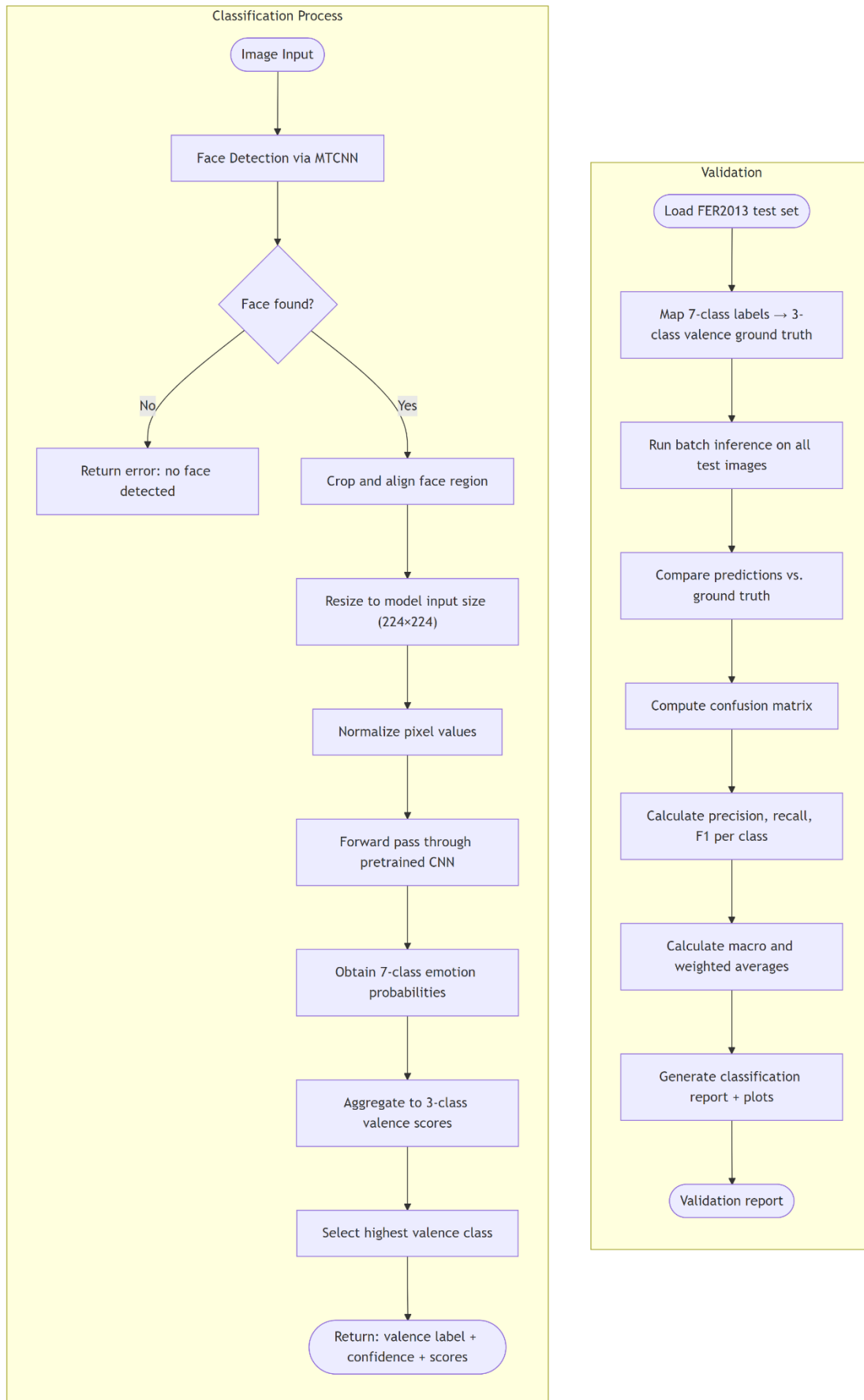


Figure 3: Process flow expressed as a UML activity diagram